

# ALFIAN NUR USY Aid

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## SUMMARY

**Computer Science** Graduate from STT Terpadu Nurul Fikri (GPA: 3.94) based in Bogor, specializing in scalable software engineering with deep mastery in **OOP**, **SOLID principles**, and **Clean Code** architectures. Experienced in building robust, high-performance web systems and RESTful APIs using **PHP (Laravel)**, **Python (Flask)**, **JavaScript/TypeScript (React, Next.js, Node.js)**, and relational database optimization with **PostgreSQL & MySQL**. Proven track record in academic and faculty-backed research projects, testable backend engineering, and mentorship as a university Teaching Assistant.

## TECHNICAL EXPERTISE

- **Core & Architecture:** Object-Oriented Programming (OOP), SOLID Principles, Clean Code, MVC Architecture, Atomic Design Pattern, Component-Driven Development, RESTful API Design, Client-Server Decoupling.
- **Languages:** JavaScript, PHP, Python, TypeScript, SQL (PostgreSQL, MySQL), HTML5, CSS3.
- **Framework & Libraries:** Laravel, Flask, Express.js, Node.js, Next.js, React, Tailwind CSS.
- **Tools & Environment:** Git/GitHub, Postman, Linux CLI, Relational Schema Design.

## PROFESSIONAL & ACADEMIC EXPERIENCE

**Laboratory Teaching Assistant (Backend, Database, & DSA)** | STT Terpadu Nurul Fikri Sep 2024 - Jul 2026

- Facilitated laboratory sessions and mentored students across Backend Engineering (PHP/Express.js), RDBMS (MySQL), and Data Structures & Algorithms.
- Evaluated practicum code repositories and provided 1-on-1 guidance on OOP principles, SQL query optimization, and complex algorithmic logic.

**Frontend & Blockchain Researcher** | Project: [DigiArc](#) Aug 2025 - Sept 2025

- Researched and built decentralized file storage architectures (Web3 Storage) utilizing Next.js, Wagmi, and IPFS data integrity protocols.
- Integrated Solidity smart contracts for cryptographically verified proof-of-storage, adhering to clean code conventions and security standards.

**Fullstack Developer (Academic Research)** | Project: [ESAO](#) Feb 2025 - Jul 2025

- Engineered ESAO (Essay Analytic Online), an AI-driven automated essay grading platform empowering lecturers to assess essay exams in seconds.
- Architected a decoupled full-stack system utilizing **Next.js** for a responsive analytics dashboard and **Flask (Python)** with **LangChain** for rubric-based, automated student feedback.

**Software Engineering Participant** | Magang dan Studi Independen Bersertifikat ([MSIB](#)) - PT Global Investment Sep 2024 - Dec 2024  
Institusi (Learning X Academy) | Project : [Chicken Yasaka](#)

- Participated in an intensive technical upskilling framework, culminating in the development of "Chicken Yasaka", a full-stack e-commerce platform utilizing Python (Flask), jQuery (AJAX), and MongoDB.
- Developed robust business logic including secure user authentication, dynamic cart management, and an admin dashboard, culminating in multiple position offers from technical partners.

## SOFTWARE ENGINEERING PROJECTS

**RoadSense (Smart GIS Navigation)** | T3 Stack, Next.js 15, tRPC, Flask, Leaflet, OSRM

- Engineered a web-based Geographic Information System (GIS) application to map road damage through crowdsourced data visualization.
- Implemented complex smart routing algorithms by decoupling the architecture, utilizing Flask (Shapely) and OSRM (Open Source Routing Machine) for high-performance geospatial calculations.

**HRD RESTful API Dashboard** | Express.js, Node.js, MySQL, OOP

- Engineered a modular and secure backend architecture for Human Resources management utilizing Express.js and MySQL.
- Implemented strict MVC patterns with class-based controllers and Data Access Object (DAO) models, ensuring data security through parameterized queries and adherence to RESTful API standards.

**L-Movie (Movie Discovery Platform)** | React, TailwindCSS, Axios

- Developed a dynamic frontend web application for real-time movie discovery and catalog exploration.
- Implemented asynchronous client-side REST API fetching to seamlessly consume and render external database catalogs, ensuring responsive UI and optimal user experience.

**Event Management System** | PHP, Laravel (MVC), MySQL

- Developed a comprehensive event registration and management platform applying strict MVC architectural patterns.
- Built secure authentication flows, robust role management, and full CRUD operations to ensure scalable data handling for event organizers and attendees.

## SELECTED CERTIFICATIONS

- **The Complete Web Development Bootcamp** – Angela Yu, Udemy | [Credential Link](#)
  - Skills: Full-Stack Web Development, PostgreSQL, Node.js REST APIs, Relational Database Modeling.
- **Complete JavaScript & Modern Backend Engineering** – Hitesh Choudhary, Udemy | [Credential Link](#)
  - Skills: Core JS Architecture, OOP Design Patterns, Asynchronous Workflows, Clean Code.
- **Certified Competence Full-Stack Web Development** – LearningX Academy | [Credential Link](#)
  - Skills: Full-Stack Application Lifecycle, Clean Architecture, Database Management.

## EDUCATION

**Bachelor of Computer Science at STT Terpadu Nurul Fikri** Jul 2022 - Jul 2026

- **GPA:** 3.94/4.00.
- **Relevant Coursework:** Object-Oriented Programming (OOP), Database Systems, Web Programming, Data Structures & Algorithms.
- **Publication:** Authored and submitted a machine learning prediction model manuscript to the *Multimedia Artificial Intelligent Networking Database Journal* (DOI: <https://doi.org/10.26760/mindjournal.v11i1.115-126>).